

Automated NLP Pipeline for Conflict Events Extraction From Multilingual African Radio Streams

Keywords

Natural Language Processing (NLP), Speech-to-Text (STT), Named Entity Recognition (NER), Large Language Models (LLMs), Event Extraction, Low-Resource Languages, Armed Conflict Data, Radio Broadcast Media

Abstract

This report introduces an automated end-to-end natural language processing (NLP) architecture designed for the continuous, real-time extraction of structured armed conflict and political violence events from multilingual African radio broadcasts. While global conflict observatories like the Armed Conflict Location & Event Data Project (ACLED) traditionally rely on human researchers mining written text, crucial localized events in areas with limited digital penetration remain trapped in auditory mediums like radio. To address this infrastructure gap, this pipeline ingests raw internet audio streams, applies Silero Voice Activity Detection (VAD), identifies regional languages via Meta's MMS-LID, and routes audio to language-optimized Speech-to-Text (STT) engines (Sunbird AI and Meta MMS-1B-All). Transcripts are diarized via PyAnnote, filtered for semantic relevance via Groq-hosted Llama 3.3 (70B), translated into English using Google Translate API, and parsed using a zero-shot Generalist Language-agnostic NER (GLiNER) model before final structural event synthesis with Llama 3.3 (70B). Evaluated over a continuous 29-day deployment period, monitoring stations across the Democratic Republic of the Congo (DRC), Uganda, and Ethiopia, the pipeline successfully processed hours of continuous radio streams. The pipeline successfully captured 157 verified relevant conflict events (of which only 33 were already documented within the ACLED database during the same period) demonstrating that local radio broadcasts contain a wealth of conflict data missed by traditional text monitoring. Ultimately, this project proves that automated NLP systems can effectively assist human analysts in expanding global conflict databases.

1 Introduction

Radio remains the primary medium for news and local communication across much of the African continent. While global conflict observatories aggregate large amounts of data from written text, localized reporting in specific regions relies heavily on auditory mediums, specifically radio.

To investigate the viability of audio-based sources events extraction, this project focuses on three distinct key countries:

1. **Democratic Republic of the Congo (DRC)**

In the DRC, internet penetration is highly restricted at roughly 30%. While formal government news is delivered in French (spoken by 51% nationwide), critical tactical and regional information is broadcasted in localized languages: Lingala (59% in the capital/west and heavily tied to the military and police forces) and Swahili (35-40% in the east). Critically, over 80% of conflict events occur within these eastern, Swahili-speaking zones.

2. **Ethiopia**

Approximately 76% to 80% of the Ethiopian population resides in rural areas, relying almost exclusively on radio networks due to routine digital disruptions and YouTube censorship. Linguistically, the primary nationwide language is Amharic (30%), supplemented regionally by Oromo (35%) and Tigrinya (6%). Consequently, processing Ethiopian audio requires specialized NLP pipelines capable of handling text outputs structured in UTF-8 Ge'ez characters (e.g., አማርኛ).

3. **Uganda**

In Uganda, radio is the dominant medium for daily news distribution, especially in rural communities. While English (50–60%) and Swahili (30–40%, heavily used by security forces) serve as official languages, Luganda is the most widely spoken indigenous language, representing 17% to 18% in crucial central economic and political hubs.

These countries were identified due to recommendations from ACLED's Global Data Collection team, and because their information environments favor sources like Radio or Television. It is more likely that such countries would have locally sourced information mentioned in Radio channels but not found in open-source written platforms covered by ACLED.

1.1 Target Baseline Metrics

The following table outlines the baseline media landscape and the specific radio streams selected for automated monitoring within each target country.

Country	Radio Penetration %	Radios Monitored	Languages monitored
DRC	52% [1]	Top Congo FM, Radio Okapi	Lingala, French
Uganda	68% [2]	Akaboozi FM, Pearl FM	Luganda, Swahili
Ethiopia	71% [3]	DW Amharic	Amharic, Oromo

Across these monitored stations, Top Congo FM, Radio Okapi, and DW Amharic are already recognized ACLED sources (though limited strictly to their text-based web news), while Akaboozi FM and Pearl FM are entirely unmonitored by ACLED.

2 Problem Definition

ACLED's primary operational mechanism relies on human researchers aggregating and coding geopolitical conflict data from global textual news sources. However, a massive amount of crucial, localized news data in African nations, specifically within Ethiopia, Uganda, and the Democratic Republic of the Congo (DRC), is broadcasted via auditory mediums like local internet radio and YouTube, rather than written text.

Audio-based reporting has remained outside the scope of ACLED's researchers dataset aggregation as extracting structured events from such sources, by physically listening to infinite hours of continuous radio streams, is incredibly daunting and time-consuming for human coders. Recent advancements in Large Language Models (LLMs) and regional Speech-to-Text (STT) models have introduced the possibility of automated media monitoring to solve this problem. This project investigates whether these audio streams could work as a new, viable data source to supplement ACLED's existing text-based datasets.

3 Research Questions

To evaluate the viability and analytical value of this automated pipeline, this study addresses the following research questions:

1. How do the events extracted from radio broadcasts compare with text-based events currently coded in ACLED's database, and are there significant differences in terms of event coverage, granularity, or timeliness?
2. How effectively can automated Natural Language Processing (NLP) architectures extract viable, geopolitical armed conflict events from continuous African live radio streams?
3. What types of conflict-related events are most frequently discussed in radio broadcasts across the three selected countries, and how do these patterns vary between countries?

4 Methodology Overview

This section outlines the architectural design of the automated pipeline developed to extract structured conflict and political event data from live African radio broadcasts. The system was designed to operate continuously, processing audio streams in real-time and produce structured event records suitable for integration into the Armed Conflict Location & Event Data Project (ACLED) database.

The pipeline contains six sequential phases: Live Stream Ingestion and Voice Activity Detection, Language Identification, Language-Aware Routing and Speech to Text (STT), Speaker Diarization and Full Transcript Construction, Relevance Classification, Translation, Named entity Recognition (NER), and Event Structuring.

The overall architecture was implemented in Python using libraries including PyTorch, Hugging Face Transformers, PyAnnote, GLiNER, and Groq-hosted Llama models. The complete implementation and replication code can be reviewed in the [Project's GitHub Repository](#).

A comprehensive visual schematic of this operational flow is illustrated in Figure 1. For a more detailed analysis of each individual processing phase, **see section 8 (Detailed Pipeline Methodology)**.

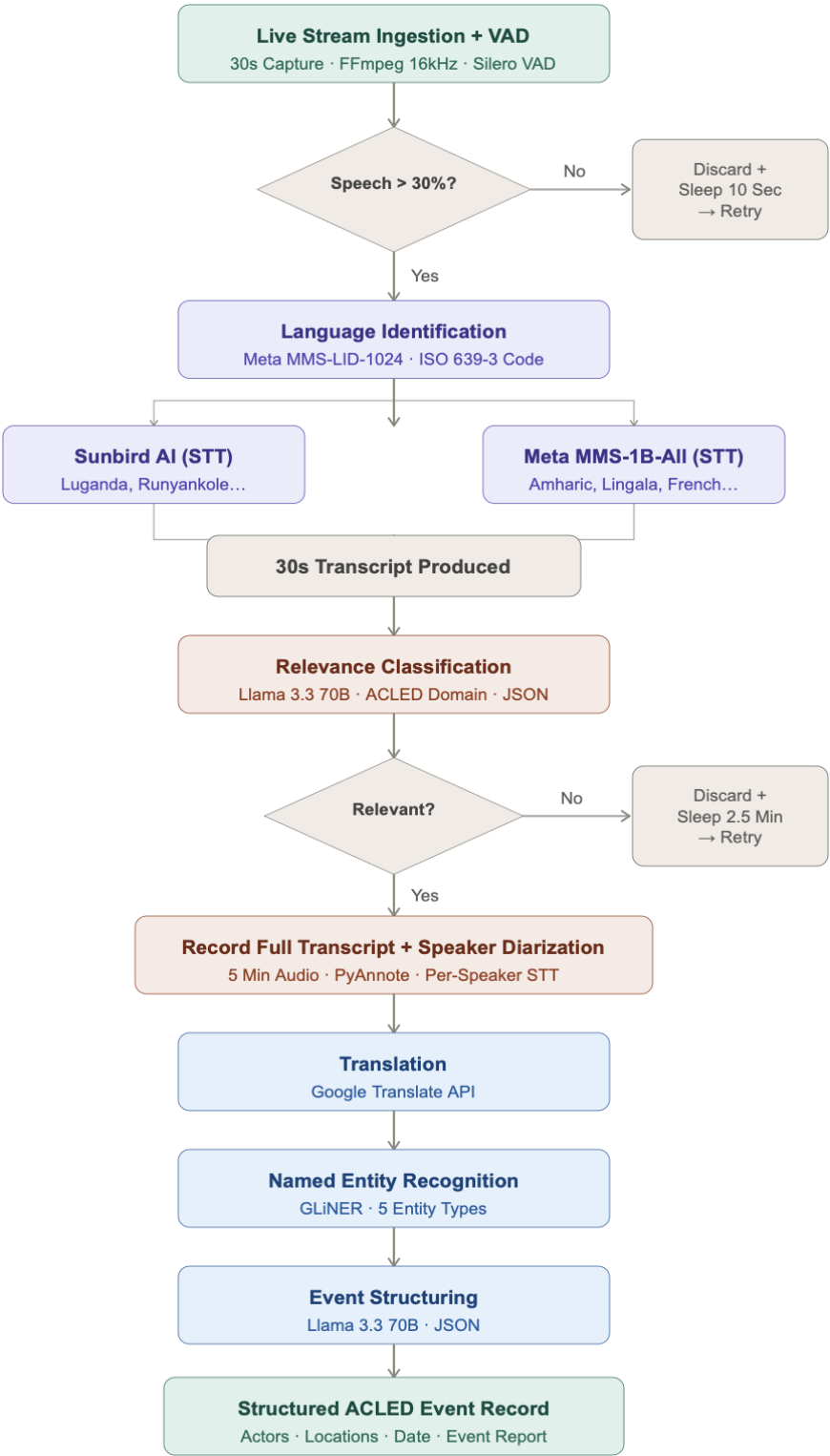


Fig 1: Pipeline Architecture Flowchart

5 Results

During the active testing and monitoring phase, which took place from 26-04-2026 to 25-05-2026, totaling 29 days, the pipeline demonstrated the ability to operate continuously and capture specific relevant data. The pipeline successfully processed continuous streams, isolating discrete 5-minute event windows from hours of raw audio.

In addition to the three targeted countries (DRC, Uganda, Ethiopia), the pipeline extracted events from four additional African countries (Nigeria, Mali, Chad, and Kenya) via DW Amharic radio stream, which broadcasts global news rather than strictly domestic Ethiopian news.

The automated pipeline achieved its highest monitoring success in Uganda and the DRC. In Uganda, this was facilitated by Sunbird AI, which provided highly accurate, language-specific transcriptions. In the DRC, success was driven by the abundance of dedicated national news stations like Top Congo FM. Ethiopia, however, was significantly harder to monitor; the Radio-Browser API yielded only a single viable stream, which broadcasted international news rather than the actionable, localized domestic reporting found in the other target countries.

5.1 Event Extraction and ACLED Baseline Alignment by country

The following table details the comprehensive distribution of overall raw and relevant system yields extracted by the automated pipeline across both target and adjacent regional conflict countries, validated directly against ACLED's human-coded records over the same monitoring period.

Country	Total Events Extracted	No. of Relevant Events Extracted ¹	No. of Events Recorded in ACLED	% of Relevance	% of Relevant Events Already in ACLED
DRC	812	119	28	14.7%	23.5%
Uganda	246	18	0	7.3%	0
Ethiopia	6	1	0	16.7%	0
Nigeria	24	7	4	29.1%	57.1%
Mali	8	5	1	62.5%	20%
Chad	6	4	0	66.7%	0
Kenya	4	3	0	75%	0

The types of events extracted ranged from political & government news to armed conflict & military developments.

¹ Relevance was assessed according to ACLED's general methodology, but they were not reviewed by regional desks.

5.2 Qualitative Comparison: ACLED's vs. Our Pipeline's Database

The following table highlights examples of identical geopolitical events captured simultaneously by human researchers (ACLED) and the automated audio pipeline.

ACLED's Database (Text-Sourced)	Our Pipeline's Database (Audio-Sourced)	Our Pipeline's Recorded event_date
"On 1 May 2026, Wazalendo attacked an MSF convoy that was traveling south to Mwenga, Kamituga and Kitutu to 'acts of aggression' on the road in Kilomoto, a 'few kilometers from Kalambi' (Luindi, Mwenga, Sud-Kivu). MSF said none of their staff was injured in the incident."	"A security incident occurred on May 1, 2026, involving a maidecine sans frontières (MSF) team traveling between Mwenga and Kamitugaki tutou. The team was immobilized at a checkpoint manned by wazalendo elements, despite prior notification to competent authorities. The situation deteriorated when armed men opened fire, and the team was forced to continue on their way amidst strong tension."	01-05-2026
"On 7 May 2026, prisoners attempted to break down a wall within the central prison in Isiro (Rungu, Haut-Uele). Prison guards fired warning shots. There were no fatalities."	"An attempted escape at Siro prison was foiled by the police and security forces. The inmates had demanded to be placed in the same cell as nine recently arrested Chinese subjects, but the management of the prison rejected the request. The situation quickly degenerated, prompting the police to fire warning shots to disperse the demonstrators."	08-05-2026
"On 5 May 2026, overnight, ADF attacked civilians in the agricultural zones around Bakaeko, specifically in Manzumbu, Musangwa, Wasafi and Mangambo, coded to Bakaeko (Babila-Babombi, Mambasa, Ituri), killing 24-26 people, mainly farmers. After the attack ADF retreated in the surrounding forest. IS claimed the attack on 8 May via Telegram."	"The ADF and allied terrorists carried out multiple incursions and attacks in the Babilababombi chiefdom, Mambassa territory, and other areas, resulting in the deaths of 24 civilians in Biacato and significant material damage. At least 23 villages were almost emptied of their inhabitants following the attacks."	06-05-2026
"On 22 May 2026, overnight, young people burned Ebola isolation tents"	"Tents built for the re-isolation of debola patients were set on fire by inconis at"	20-05-2026

set up by MSF in Mongbwalu (Djugu, Ituri). 13-18 Ebola patients fled. No one was inside or injured.”	the general hospital of mongwalo, with references to very sick debola patients trying to escape.”	
“Property destruction: On 15 May 2026, suspected JNIM intercepted a vehicle transporting livestock traveling to Bamako in Sirakorola (Koulikoro, Koulikoro), offloaded the cattle, and burned the vehicle.”	“The support group for Islam and Muslims, led by Jim, blockaded the main roads leading to Bamako, and dozens of transport vehicles and goods heading to the Malian capital were set on fire by dihadists.”	12-05-2026

5.3 Events Absent from the Baseline ACLED Database

The following table shows examples of critical localized events successfully extracted by the automated pipeline that were completely absent from ACLED's text-based database during the same monitoring period.

Country	Recorded Event	Recorded event date
Uganda	“Police participated in the torture of a 10-year-old boy at Nkoowa village in Wakiso district. The boy was tied to a window with a rope after being accused of stealing. The incident occurred in the evening, around 10 o'clock.”	07-05-2026
DRC	“Armed bandits attacked a maternity ward in Diinjili, killing a newborn and injuring a mother.”	04-05-2026
DRC	“The Armed Forces of the Democratic Republic of Congo, through the thirtieth main defense brigade, have reconquered strategic localities in the territories of Mambassa, allowing for the restoration of state authority and the free movement of people and goods. The operations also led to the reopening of a strategic axis connecting the areas.”	12-05-2026
DRC	“Thirteen people were arrested in Kinjasa Kimbanzeke for burglarizing and desecrating the wall santéopil and the church of Kriso Congo. The police, led by Israel Kantuf and Brown Captain Munongo, dismantled the gang, which included a military police officer.”	08-05-2026
DRC	“An armed robbery occurred in the Honnel district of Kasavubu, resulting in the theft of 130 million Kongolese francs and the death of one person. Thirteen others were also affected.”	08-05-2026

Kenya	“At least seven hundred and ten people were arrested on Monday during protests and a public transport strike against the increase in fuel prices in Kenya. The police reported four deaths and thirty injuries.”	16-05-2026
Mali	“At least 25 people were reportedly killed in coordinated jihadist attacks targeting several villages in the Bandia Kara region. The attacks caused numerous injuries, houses were set on fire, and livestock were taken away, leading to the flight of several families. The majority of victims were civilians.”	20-05-2026
Chad	“The government arrested the former leader of the opposition and former prime minister, Susai Masra. International human rights organizations, including the Women's Rights Ward and Firedom House, have described this incident as a dictatorship.”	08-05-2026
Nigeria	“A group of students were kidnapped from a school in Ugamasho town by the Bundigas. The state police and security forces are currently in talks to rescue the students. Seven students and seven teachers from Skandre are reported to have been kidnapped, with some teachers killed. The kidnappers used a jungle between the south and the north, especially the agwamaso ajia market.”	19-05-2026

5.4 Thematic Trends by Country

Qualitative analysis of the extracted event database reveals distinct thematic trends that reflect the unique geopolitical realities and information on environments of each target country.

In the Democratic Republic of the Congo (DRC), the extracted data is dominated by reports of armed violence, detailing clashes between the national military (FARDC), local defense militias (Wazalendo), and foreign rebel factions (such as the ADF and M23). Beyond direct military engagements, the streams frequently captured severe violence against civilians, mass displacements, and civil unrest reacting to regional insecurity.

In contrast, Ugandan broadcasts entirely lacked reports of armed combat, focusing instead on institutional governance, political friction, and localized crime. The conflict architecture extracted here is defined by state-driven actions, including crackdowns on political opposition, the detention of prominent figures, and police disputes at party offices. Additionally, Ugandan radio places a unique emphasis on high-profile financial fraud, central bank robberies, and isolated incidents of police misconduct.

Overall, the data illustrates a clear geographic shift in the nature of conflict reporting. While the DRC information ecosystem uses radio to crowdsource survival metrics against active warfare, Uganda utilizes the medium to track civic disputes, police actions, and economic crime.

6 Limitations

Evaluating the performance of the automated radio monitoring pipeline showed several critical technical, linguistic, and structural constraints that impact its scalability.

6.1 Source Discovery Constraints

The structural diversity of the monitored stations is entirely bounded by the public directory listings available on the *Radio-Browser API*, meaning that if a highly critical regional conflict station does not maintain an active, valid stream listed within this open repository, it remains entirely invisible to the pipeline's exploration architecture. Furthermore, the architecture is structurally limited to radio stations that broadcast live streams over the web. Traditional terrestrial-only AM/FM frequencies, which often carry highly localized, rural conflict reporting, cannot be ingested without specialized local hardware, such as dedicated physical antennas, transceivers, and specialized hardware capture equipment.

6.2 Platforms Anti-Scraping and Token Discontinuity

While YouTube and similar video hosting networks act as major vehicles for regional African news distribution, their API terms of service explicitly forbid downloading, scraping, or permanently caching raw videos. Consequently, implementing continuous automated terminal tools like yt-dlp results in aggressive IP-address flagging and systematic anti-bot blocking over prolonged running. Additionally, the vast majority of standard stream broadcasts host their digital audio feeds via volatile proxy configurations that utilize temporary, token-based signature URLs. These forcefully expire every few hours, breaking hardcoded pipeline connections with 403 Forbidden credential errors. Compounding this issue, aggressive anti-bot tracking modules deployed by streaming distribution networks (such as Zeno.fm) actively blacklist IP address ranges belonging to public cloud environments (e.g., Google Colab). When an automated server connection is detected without a valid browser handshake, the host forcefully drops the connection, which results in a network drop or read timeout.

6.3 Spatiotemporal Context Windows

The primary stream gating mechanism relies entirely on an initial 30-second audio context sample to evaluate speech activity and semantic relevance. This small window introduces a sampling bias; if the sentinel loop tests a stream during a brief station transition, a slow introduction, or a prolonged pause, the gating model will mistakenly tag the broadcast as irrelevant noise. As a result, the system will discard the frame and trigger a sleep state, potentially missing a critical conflict event.

6.4 Low-Resource Linguistic Scarcity in STT Frameworks

Because indigenous African languages are classified as low-resource languages in NLP research, available Speech-to-Text models lack exhaustive corpus validation. This scarcity drives frequent model hallucinations when encountering regional background noise or overlapping voices. Furthermore, the downstream transcription engines translate names and locations phonetically (how they sound to the model) rather than orthographically (how they are spelled in dictionary gazetteers). For instance, the system may generate acoustic variations like "Kamitugaki" when referencing the actual location "Kamituga." This discrepancy introduces

transcription friction that can cause downstream Named Entity Recognition (NER) models to overlook or misclassify critical geographic variables.

The transcripts are also concatenated and not punctuated because of the phonetic sequence mapping anomalies inherent to low-resource transcription systems.

6.5 Third-Party API Interdependence and Rate Limiting

The pipeline's systemic uptime is fundamentally restricted by the operational availability of external API providers. High-parameter Large Language Models (LLMs) enforce rigid Tokens Per Day (TPD) and Requests Per Minute (RPM) constraints, which strictly govern how frequently the pipeline can wake up to poll the audio landscape without throwing rate-limit crashes. Furthermore, heavy server traffic on specialized regional STT endpoints (Sunbird AI) often results in network timeouts that delay real-time extraction metrics.

6.6 Technical Architecture vs. Human Analysts

Fully automated systems lack the contextual flexibility to adapt to unstructured changes in broadcasting culture. For example, if a target station experiences a connection drop or shifts to playing continuous music blocks for hours, an unmonitored pipeline will continuously loop, running expensive, redundant GPU cycles on music or dead air. Additionally, while the combined LLM-NER pipeline is exceptionally skilled at synthesizing raw audio into clear text schemas, it cannot completely simulate the nuanced situational judgment of an intelligence analyst. The system frequently extracts general historical facts, administrative legal updates, or duplicate event windows that meet structural conditions but fail true geopolitical relevance, ultimately requiring human analysts to filter and clean the final data repository before integration into databases.

7 Recommendations and Conclusion

In conclusion, the testing of this automated NLP pipeline confirms that radio broadcasts represent a critical, untapped reservoir of geopolitical conflict data. As demonstrated by the significant volume of relevant events successfully extracted by the pipeline that were entirely absent from ACLED's text-based database, traditional open-source monitoring possesses inherent blind spots in regions heavily reliant on auditory mediums. Consequently, it is highly useful to widen the scope of media monitoring to systematically include regional radio streams in targeted countries to capture these localized security alerts.

To scale this automated audio monitoring pipeline from a prototype into an enterprise-level product, several strategic and infrastructure upgrades are recommended.

7.1 Partnerships and API Upgrades

Establishing formal data-sharing relationships or direct internal feeds with major independent broadcasters (e.g., Akaboozi FM) can bypass internet stream fragmentation, proxy drops, and anti-bot blocking at the point of ingestion, also this way we will have more stations rather than being limited by what is there in Radio-Browser API only. Additionally, because the pipeline's reliance on free-tier APIs introduces structural vulnerabilities regarding daily usage caps, it is advised to secure a direct production agreement with Sunbird AI. To maximize endpoint stability

across the text-processing components, a corresponding enterprise production tier should be secured for the Llama 3.3 foundational models via Groq. This would raise the structural Tokens Per Day (TPD) and Requests Per Minute (RPM) limits, mitigating the system rate-limit crashes noted during sudden, text-dense streaming surges

7.2 Meta MMS Deployment

Because Meta's MMS-1B-All framework requires continuous, heavy GPU compute power for speech recognition and does not offer a public cloud API endpoint, two viable deployment paths are recommended for production. The first option is to rent a cloud instance, such as an AWS EC2 g5.xlarge with an NVIDIA A10G 24GB GPU. While this adds an operational cost of approximately \$0.42 to \$0.50 per hour, it allows for dynamic scaling during peak conflict periods. Alternatively, a local setup can be established by purchasing a dedicated workstation laptop equipped with a high-end consumer GPU, like the NVIDIA RTX 4090. This requires a one-time capital expense of roughly \$1,500 but eliminates ongoing hourly cloud processing fees, enabling the model to run 24/7 indefinitely.

7.3 Human-in-the-Loop Integration

The pipeline must function as a high-velocity automated filter rather than an autonomous publisher. Generated event logs should route to an intermediate staging dashboard, where human monitors and intelligence analysts review the transcripts, filter out rhetorical political noise, and validate entries before integration into ACLED's public database.

7.4 Domain-Specific Fine-Tuning on ACLED Methodology

While the current pipeline relies on prompting generalized Large Language Models to structure events, these models inherently lack an intrinsic understanding of the specialized coding rulebook utilized by the Armed Conflict Location & Event Data Project (ACLED). A critical recommendation for future scalability is to perform instruction fine-tuning on an open-source foundational model using ACLED's extensive historical database and official methodology guidelines. This way, the system would shift from relying on prompt engineering to possessing a native, internal comprehension of ACLED's strict taxonomies, actor definitions, and geopolitical event thresholds. This domain-specific training would significantly reduce the rate of irrelevant extractions and ensure the generated schemas are immediately compliant with ACLED's database standards prior to human review.

8 Detailed Pipeline Methodology

This section provides detailed analysis and documentation of the technical components and algorithmic configurations driving the automated radio extraction pipeline.

8.1 Radio Stations Sourcing

Before the pipeline begins, querying the Radio-Browser public API (Radio-Browser is an open-source, community-driven directory of global radio broadcasts) to retrieve a broad catalog of online radio stations broadcasting from the three target countries: Ethiopia (ET), Uganda (UG), and the Democratic Republic of Congo (CD) is performed. The query is constrained to

stations tagged as ‘news’ broadcasts and filters out broken stream URLs. The resulting station list is then refined locally using a keyword-matching heuristic applied to both station names and metadata tags, retaining only stations plausibly associated with news or current affairs programming (e.g., terms such as *news*, *info*, *actualite*, and *zena*). This two-stage approach ensures coverage while reducing noise from entertainment and music streams. This step was done once only to source viable radio stations suitable for live stream ingestion.

The list of chosen stations after this step includes: Top Congo FM and Radio Okapi (DRC), Akaboozi FM and Pearl FM (Uganda), DW Amharic (Ethiopia).

8.2 Live Stream Ingestion and Voice Activity Detection

For each given stream URL, the pipeline captures a 30-second audio sample via HTTP streaming. The raw audio is then pre-processed using FFmpeg, which resamples the file to 16 kHz mono WAV format to ensure compatibility with downstream models. The cleaned sample is passed to the *Silero Voice Activity Detection (VAD)* model to determine whether the recording contains human speech. The ratio of speech frames to total frames is computed, and only recordings exceeding a 30% speech threshold are retained for further processing; recordings identified as music or dead air are discarded. This step prevents unnecessary computation on non-informative content.

8.3 Language Identification

Retained audio samples are passed to Meta's *MMS-LID-1024*, a multilingual language identification model trained across over 1,000 languages. The model processes up to 30 seconds of audio and outputs a predicted ISO 639-3 language code. This code is then used to select the appropriate speech-to-text (STT) model in the next stage. The full language name is resolved from the ISO code using the *pycountry* library for logging and reporting purposes.

8.4 Language-Aware Routing and STT

Given the linguistic diversity of the target regions, spanning Luganda, Runyankole, Oromo, Lingala, Swahili, French, and Hausa, a single STT model cannot serve all streams since different models perform better for certain languages. After testing, we found out that Sunbird AI works better for Ugandan languages while Meta MMS is more suited for DRC and Ethiopia Languages. The pipeline therefore implements a routing function that dispatches audio to one of two models based on the detected language code.

For Ugandan languages (Luganda, Runyankole,), the pipeline routes to *Sunbird AI's specialized API*, which provides dedicated adapter models for Ugandan languages. For languages associated with the DRC and Ethiopia (Amharic, Oromo, Lingala, Swahili, and French) the pipeline uses *Meta's MMS-1B-AI*, a multilingual Wav2Vec2-based Connectionist Temporal Classification (CTC) model that supports over 1,100 languages via swappable language adapters. The adapter corresponding to the detected language is loaded at inference time, and the model output is decoded via greedy CTC decoding.

Those models were selected, rather than the standard Open AI's Whisper or Google's Speech to Text, because they support low resources African languages.

The accuracy of the STT models depends on factors such as how slowly and clearly the speech is pronounced. For example, speech is more easily recognized during news broadcasts, where

presenters are trained to articulate clearly, and is less recognized during call-ins, where the quality of the audio is poorer and the speech is a rapid conversational style.

8.5 Speaker Diarization and Full Transcript Construction

When an initial transcription is classified as relevant (described in Section 8.3), the pipeline records a longer segment (five minutes, captured as ten sequential 30-second chunks) for deep analysis. The merged audio is converted to 16 kHz mono PCM WAV format and submitted to the *PyAnnote Speaker Diarization Community model (v1)*, which segments the audio into speaker turns and assigns speaker labels.

The raw diarization output is post-processed by a segment merging step that collapses consecutive utterances from the same speaker separated by less than two seconds which reduces over-segmentation caused by natural pauses and breath gaps. Segments shorter than 1.5 seconds are discarded as likely artifacts (coughs, interjections, etc.). Each surviving speaker turn is extracted as an individual audio chunk using FFmpeg and routed through the same language-aware STT system described in Section 8.4. The resulting per-speaker transcripts are stitched together with speaker labels (e.g., [SPEAKER_00]: ...) to produce a full, attributed broadcast transcript.

8.6 Relevance Classification

Before committing resources to full-length recording and further NLP processing, the pipeline performs a relevance gate on the initial 30-second transcript. The transcript is submitted to *Llama 3.3 (70B)* via the *Groq inference API* with a structured prompt that instructs the model to identify the primary topic and classify it as either relevant or irrelevant according to ACLED's domain of interest (political and conflict news).

Relevant categories include political developments, armed conflict, rebel activity, protests, civil unrest, state violence...etc. Strictly irrelevant categories (including disease outbreaks, natural disasters, sports, entertainment, weather, and religious content) trigger automatic rejection. The model responds in structured JSON format, providing both a binary relevance flag and a one-sentence justification. Irrelevant broadcasts trigger a 2.5-minute sleep (this was chosen based on trials, initially it was 5 mins but that would result in missing a lot of useful info, and then we tried 1 min but that resulted in running out of API limit per day) before the next sampling cycle, preventing unnecessary API calls during non-news programming blocks while also not missing a lot of relevant news when they start a new block.

8.7 Translation, Named Entity Recognition, and Event Structuring

Full transcripts passing the relevance gate are processed through a three-step NLP pipeline. First, the raw text is cleaned to remove noise artifacts (repeated characters, garbled tokens, and excessive whitespace) and segmented into 250-word chunks due to models input constraints. Each chunk is translated into English using the Google Translate API via the deep-translator library, with automatic source language detection.

The pipeline faced two architectural options: Translate transcripts into English first then apply NER, or clean the transcripts and apply multilingual NER directly in the African languages.

8.8 Comparing the two NLP Architecture Approaches

The following table outlines the advantages and disadvantages encountered when deciding where translation should occur within the data processing pipeline.

Approach	Pros	Cons
Clean -> Translate -> NER	Easier to evaluate against ACLED's English data; easier debugging; English NER models are significantly more accurate.	Translation may distort local names.
Clean -> Multilingual NER	No translation distortion.	LLM cleaning can be unstable; debugging is exceedingly difficult without fluency in local dialects.

The first approach (Translation before NER) was selected. The Google Translate API naturally resolves formatting issues (lack of punctuation and concatenated words) via built-in cleaning algorithms, outperforming alternatives like Meta's NLLB-200 in our testing.

Named entity recognition (NER) is then performed on the English-translated text using GLiNER (Generalist Language-agnostic NER), a span-based extraction model fine-tuned on diverse entity types. Five entity classes (labels) are extracted: person, organization, location, date, and event. Duplicate entities across chunks are resolved by retaining the instance with the highest confidence score. Extracted date strings are normalized to ISO 8601 format (DD-MM-YYYY) using dateparser. GLiNER was chosen, rather than NLTK or spaCy, for its flexible zero-shot, and the ability to define our own labels dynamically.

Finally, the translated transcript and the extracted entity list are jointly submitted to *Llama 3.3 (70B)* with a structured prompt that instructs the model to group the entities into discrete conflict or political events following ACLED's coding methodology (not entirely).

The model is required to produce a JSON object containing, for each event: an event name, a list of actors involved, a list of locations, a date (calculated from relative references against the extraction date, or marked as Extraction Date 'ED tag' if absent), and a summarized event report detailing specific tactics, casualties, and targets where mentioned. Temperature is set to zero to maximize output determinism.

8.9 Output and Storage

Structured outputs are saved to Google Drive in a country-stratified directory. For each confirmed event, three files are produced: the cleaned WAV audio recording, the raw diarized transcript, and a formatted event report containing all pipeline outputs (transcript, translation, NER table, and structured event records). Files are named using a {country}_{type}_{index} convention (e.g., uganda_event_3.txt), and a persistent session log records all pipeline decisions with timestamps.

8.10 Filtering and Cross Checking

The extracted events are then extracted from the files and grouped to an excel file and then filtered to discard irrelevant events, the relevant events are then cross-checked against

ACLED's data for the same time period. The irrelevant events discarded included general facts, events lacking details, events that are irrelevant to ACLED's domain and events that are about international news (eg: Iran and Israel war), and events that are historical or about old events that are way out of the monitoring period.

References

- [1] Target SARL. (2024). *WRD 2024: Impact of Radio in the Democratic Republic of Congo*.
<https://target-sarl.cd/en/?read&enc=utf-8&t=258&rt=WRD+2024+%3A+Impact+of+Radio+in+the+Democratic+Republic+of+Congo>
- [2] Uganda Radio Network. (2023). *Radio still leading source of information amid digital revolution*.
<https://ugandaradionetwork.net/story/radio-still-leading-source-of-information-amid-digital-revolution>
- [3] Grokipedia. (2024). *Radio in Ethiopia*.
https://grokipedia.com/page/radio_in_ethiopia